CSC291 – Software Engineering Concepts
Lecture 9 & 10: Requirements Engineering Process & Discovery Techniques

1. Requirements Engineering (RE)

Definition: The science and discipline concerned with analyzing and documenting requirements.

Goal: Create and maintain a system requirements document through four activities:

- Feasibility Study – assess if the system is worth building
- Elicitation & Analysis – discover requirements
- Specification – convert requirements into standard form
- Validation – confirm requirements match what the customer wants

2. Feasibility Study

A short, focused study early in the RE process. Answers three key questions:

- Does the system contribute to organizational objectives?
- Can it be built within schedule and budget using current technology?
- Can it integrate with existing systems?

If any answer is 'no', the project should likely not proceed.

3. Requirements Elicitation & Analysis

Also called **requirements discovery**. Engineers work with stakeholders (end-users, managers, domain experts) to uncover application needs and operational constraints.

Process Activities

- **Discovery:** Interact with stakeholders; uncover domain requirements.
- **Classification & Organization:** Group related requirements into clusters.
- **Prioritization & Negotiation:** Resolve conflicts between stakeholders.
- **Documentation:** Formally record the agreed requirements.

Common Problems

- Stakeholders don't know what they really want.
- Conflicting requirements from different stakeholders.
- Organizational/political factors distort requirements.
- Requirements change during the process.

4. Discovery Techniques

4.1 Interviews

Formal or informal conversations with stakeholders. Two types:

- **Closed:** Pre-defined set of questions.
- **Open:** No fixed agenda; wide-ranging discussion.

Good interviewers are open-minded, avoid pre-conceived ideas, and prompt discussion effectively.

4.2 User Stories

Short descriptions of functionality in customer terms: *"As a <role>, I want <goal> so that <benefit>."* Used in agile methods; contain just enough detail to estimate effort.

4.3 Scenarios

Real-life usage examples that stakeholders can relate to and critique. Each scenario includes: starting situation, normal flow, what can go wrong, concurrent activities, and end state.

LIBSYS Example: User logs in → selects article → provides payment → fills copyright form → PDF downloaded → printed. Print-only articles are deleted after printing.

4.4 Use Cases

Identify **actors** and their interactions with the system. A complete set covers all possible interactions. Sequence diagrams add event-ordering detail.

LIBSYS actors: Library User, Library Staff, Supplier. **Use cases:** Article Search, Article Printing, User Administration, Catalogue Services.

4.5 Ethnography

An **observational technique** where the analyst immerses themselves in the work environment to discover implicit requirements and understand social/organizational factors.

5. Requirements Validation

Ensures requirements define what the customer **really** wants. Fixing a requirements error after delivery can cost **up to 100×** more than fixing it during implementation.

Checks

- **Validity:** Does the system support the customer's actual needs?
- **Consistency:** Are there conflicting requirements?
- **Completeness:** Are all required functions included?
- **Realism:** Can requirements be implemented within budget?

Techniques

- **Requirements Reviews:** Manual analysis by a review team.
- **Prototyping:** Executable model to check requirements.
- **Test-case Generation:** Write tests to verify testability.

6. Requirements Management

Manages **changing requirements** throughout development. Requirements evolve as business needs and understanding change.

- **Enduring requirements:** Stable; derived from core organizational activities.
- **Volatile requirements:** Change during development or system use.

Change Management (3 Stages)

- **Problem analysis:** Discuss the problem and propose a change.
- **Change analysis & costing:** Assess impact on other requirements.
- **Change implementation:** Update the requirements document accordingly.

Key Points

- RE = feasibility study + elicitation/analysis + specification + management.
- Elicitation is iterative and involves classification, prioritization, and validation.
- Multiple stakeholders → different (sometimes conflicting) requirements.
- Social/organizational factors heavily influence requirements.
- Validation checks: validity, consistency, completeness, realism, verifiability.

Reading: Chapter 4 (Sommerville) & Chapter 5 (Pressman) | Resource: https://sites.google.com/cuilahore.edu.pk/sec